Exploring the Potential of Hackathons as a Means to Promote Understanding of AI Literacy: A Case Study

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Abstract. In this paper, we discuss the potential of using hackathons to promote critical discussions among participants from diverse backgrounds in the context of Artificial Intelligence. Our main objective for this work was to explore the viability of utilizing a one-day hackathon to bring together stakeholders, such as researchers, teachers, students and developers, to share their perspectives on AI literacy while collaboratively designing digital learning materials to promote AI literacy. In the current landscape, awareness and AI literacy have become essential. There is, however, no clear consensus on the definition of AI literacy, the needed target focus for teaching it, or how to design (digital) learning materials and curricula. Our findings show that the participants' mixed backgrounds contributed to a meaningful discussion, and the hackathon outcomes were characterized as relevant and appropriate for teaching AI Literacy. We envision that this work contributes to the discussion about reaching a consensus on the definition of AI literacy, how to design (digital) learning materials and curricula, and who to include when creating these materials, by using a participatory design approach.

Keywords: Hackathons · Computer supported cooperative work · participatory approaches · AI literacy

1 Introduction

In this paper, we explore the potential of hackathons to facilitate critical discussions about cutting-edge technologies and their societal impact. Through their collaborative nature these events can provide fertile ground for information

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exchange, knowledge building and establishing common ground between participants from diverse backgrounds and different expertise as they strive to create a shareable artifact within a limited time frame [4].

Concretely, we organized a hackathon to promote discussions about Artificial Intelligence (AI) literacy and create (digital) learning materials. AI literacy is a critical topic for the digital era, particularly as the infiltration of AI in multiple aspects of everyday life grows [2]. Several competency frameworks have emerged to support teaching and training AI literacy, such as the UNESCO frameworks [8, 9]. However, related research points out that there is no clear consensus regarding the definition of AI literacy, what the target focus in teaching AI literacy should be or how to design learning materials and curricula [2,10].

To address these challenges, we engaged stakeholders from different backgrounds – educators, AI experts and students – in a hackathon to co-create digital materials for teaching and training AI literacy. To further study how hackathons can potentially promote common understanding of AI literacy among diverse stakeholders, we formulated the following research questions:

RQ 1: How relevant were the participants' initial pitches in regard to AI literacy, and how did these evolve throughout the hackathon?

RQ 2: What are the results of the event in terms of satisfaction, learning and the participants' intention to continue on their team projects?

The innovativeness of this research lies in using an event format typically associated with rapid prototyping and technology development [1], to support and promote critical discussions among stakeholders with diverse perspectives towards establishing common ground. We envision that this work contributes to the ongoing discussion on fostering diversity and equitable access by employing a participatory design approach to build a shared understanding of AI literacy.

2 Related Work

As AI aspires to revolutionize societal functions, fostering AI literacy emerges as a priority for education [14,17]. [7] define AI Literacy as *"a set of competencies [...] to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home and in the workplace"*. AI literacy, thus, encompasses an understanding of AI technologies, their applications, and the ethical implications arising from their use [5] while occupying a pivotal intersection of digital and data literacy [7]. As such, there is a need for interdisciplinary approaches that align AI technologies with classroom practices and educational objectives [16].

To establish a common understanding of AI literacy, frameworks about AI competencies focus on identifying the skills that individuals should acquire to navigate the AI-era such as, the UNESCO AI Competency Frameworks for students [9] and teachers [8], the AICOMP framework [3] or the framework proposed by [7]. However, although these frameworks indeed define the skills that individuals should master, they do not discuss how these should be achieved. Despite

the increasing interest on the topic, there seems to be a lack of clarity regarding what should be included when teaching about AI literacy, how to teach AI literacy [15] or how to assess AI literacy [2].

We argue that fostering AI literacy requires a human-centered, interdisciplinary approach involving diverse stakeholders—such as educators, researchers, and students—in participatory co-design practices like hackathons.

Hackathons are time-bounded events and are commonly organized around a specific theme [4]. Teams are formed to work on topics within that theme and create artifacts to be shared at the end of an event, which, at best, will be continued or reused after the event [11]. To support participants, organizers commonly develop scaffolding [1], such as topic-related talks, mentoring [12], or checkpoints where teams present their progress and receive feedback [13].

We aim to design an event format where anyone, regardless of their prior knowledge, can actively participate, and that fosters the development of a shared conception of an issue or concept while retaining the quality of artifact creation.

3 Methodology

We conducted a case study of a hackathon that focused on designing materials for training AI literacy. To that end, we invited senior and early-career researchers as well as teachers and students with a background in AI in Education. The hackathon was attended by 26 participants overall who split into 4 teams during the event based on their personal interests. Except for one team (Team C), all teams had a mix of technology and education backgrounds as well as students, senior researchers, educators, developers and educational technology (Edu-Tech) professionals (overview in supplementary materials).

3.1 Event Design

The hackathon was organized as a one-day event at a major international conference for AI in Education, and was co-organized by the authors of this paper, following the guidelines of the Hackathon Planning Toolkit [1]. Figure 1 presents an outline of the event's structure and the timing of the data collection (a more detailed overview is included in the supplementary materials).

Fig. 1. Overview on event setup indicating the data collection points.

3.2 Data Collection and Analysis

To answer our research questions, we collected data through **pre- and post-questionnaires**, **artifacts** created during the hackathon – that is pitches, initial team presentations, final presentations and submitted products –, and **semi-structured interviews**; see Table 1 for an overview of the data collection methods and their relation to the research questions. A more detailed overview of the data collection methods can be found in the supplementary materials.

For the *relevance for (teaching) AI literacy* and the *aspects of AI literacy* [7] covered in the collected artifacts, two authors with expertise in AI in Education coded the collected artifacts. They first worked separately, before discussing potential disagreements and revising the codings if needed. The raters had a moderate agreement for the relevance for (teaching) AI literacy ($\kappa = 0.41$) and near perfect agreement for the covered aspects of AI literacy ($\kappa = 0.84$). We determined combined scores for relevance for AI literacy by calculating the mean scores of both ratings and used an AND logic for the addressed aspects of AI literacy (only taking into account the aspects assigned by both coders).

Table 1. Data and methods used in the analysis of the research questions

	Data (Source)	Method
RQ 1	*Relevance to... ...AI literacy*	**Artifacts**: Coding by experts on [1,5] scale
	...teaching AI literacy	**Artifacts**: Coding by experts on [1,5] scale
	Addressed AI Literacy aspects	**Artifacts**: Coding by experts following the categories proposed by Long and Magerko [7]
	prior knowledge	**Interview**: report
	AI Literacy understanding	**Pre- & Post-Quest.**: report, comparison
	impact of talks	**Interview**: report
	personal background	**Observation and interview**: report
	team work	**Interview**: report
RQ 2	*satisfaction*	**Post-Quest. and Interview**: Mean and standard deviation, report
	perceived learning	**Post-Quest. and Interview**: Mean and standard deviation, report
	intention to continue	**Post-Quest. and Interview**: Mean and standard deviation, report

4 Results

4.1 Relevance of Hackathon Projects Regarding AI Literacy (RQ 1)

All four teams followed different approaches regarding the types of materials they designed. Screenshots and details on the teams projects can be found in

the supplementary materials. Team A prototyped an app that would use an LLM to teach about AI through text, visuals, and audio. Team B created mock-ups for a web platform explaining AI concepts, differences from human intelligence, and potential issues. Team C drafted guidelines and a platform to help teachers and students integrate AI in the classroom. Team D designed a wireframe and gameplay for an online game to foster critical thinking and human agency.

Table 2. Overview on relevance to (teaching) AI literacy (on [1–5] scale) and covered AI aspects of the artifacts, that is 9 pitches and 4 teams presentations and products

Artifacts	Relevance to AI Literacy			Relevance to teaching AI Literacy			Covered aspects of AI Literacy		
	range	*M*	*SD*	range	*M*	*SD*	range	*M*	*SD*
Pitches	2–5	3.833	1.225	1–5	2.944	1.446	1–3	1.889	0.782
Initial Pres.	2–5	4.125	1.436	1.5–5	4.00	1.683	0–3	2.25	1.5
Products	3.5–5	4.125	0.629	2–5	3.875	1.315	1–4	2.5	1.291
Final Pres.	4.5–5	4.75	0.289	3.5–5	4.50	0.707	2–5	3.00	1.414

Table 2 shows the artifacts' aggregated relevance to AI literacy and training AI literacy as well as the number of covered AI aspects. The mean values increased over time, from pitches to final presentations, although the final products ranked lower than the final presentations. Similarly, the number of AI literacy aspects covered in the artifacts is rising from pitches to final presentations.

In the pre-questionnaire, 9 participants (of 25) chose the intended **definition of AI literacy** (option 1, see questionnaire in supplementary materials). The most popular answer (12 participants) was option 6. In the post-questionnaire, one person corrected their answer from option 6 to 1, and 4 maintained their previous responses (1 with option 6 and 3 with the intended option 1).

For the **items falling under AI literacy** 2 participants gave 50% correct replies, 10 participants gave 66.7% corrects replies, 6 gave 83.3% correct replies and 8 participants matched the intended response 100%. These findings indicate that even among participants who work on AI in Education, the relevant topics for AI literacy can be ambiguous.

The Interviewees report a *"more or less" (I1)* common understanding of AI literacy in their team, stating that *"[i]t all comes from different perspectives" (I2)*.

Tasks within the teams were distributed based on prior knowledge or experience: *"[members] were taking initiative to do [tasks] as they were coming up" (I1)*. *I2* – who has an educational background – explained that they joined their team because of another team member who *"also comes from the conceptual part"* and was *"ask[ing] deeper questions, not just the technical ones. [Such as] 'But what do we actually want to achieve?"'*. Regarding the work mode, they said that *"the computer scientists were discussing"* more practical aspects. While the two with educational background *"tried to initiate this discussion on the learning outcomes and the potential long-term consequences of this game [...] But I think our provocative questions were also accepted and discussed." (I2)*.

4.2 Satisfaction, Learning and Continuation Intentions (RQ 2)

In the post-questionnaire the participants *somewhat agree* with being satisfied with their projects ($M = 1.05, SD = 0.3$), which was also apparent in the interviews: *"I'm very happy with the product [... and] the dynamics of the team" (I1)* and *"This was great. And the discussions that we had were also very provocative" (I3)*. The interviews highlighted that critical discussions about AI literacy fell short: *"I am not very satisfied in terms of the discussion on the impact of AI. And it's a bit concerning that even in the event where you have the whole day, and [...] are meant to discuss things, you still have no time to do that" (I2)*.

Participants reported having learned something about AI literacy in general ($M = 1.5, SD = 0.577$), a bit about how to teach AI literacy ($M = 0.5, SD = 1.732$) and creating teaching or learning materials ($M = 0.25, SD = 1.5$). Additionally, aspects like communication, project management and game development were mentioned in the post questionnaire and the interviews: *"I learned a bit about collaboration" (I4)*. Learning was reportedly similarly inhibited by the time constraints: *"there was not enough time for me to know more [...] mainly there was time to build upon what I already knew"* (*I1*).

Participants were disinclined to continue with their projects ($M = -0.083$, $SD = 0.877$). The interviews highlighted that while participants may be interested – *"I'd love to." (I1)* – they either do not have the time or see re-connecting with their group members as a hindrance: *"I don't know if [the other team members] want to continue the idea" (I3)*. One participant took aims to *"have some kind of workshop [...] internally" (I4)* at their place of work.

5 Discussion

Concerning the relevance of the hackathon projects for AI literacy (RQ 1), the expert assessments and their agreement scores suggest that AI literacy is not yet thoroughly defined. This is also apparent in the projects' various outcomes (e.g. pitches, presentations, and deliverables). Our findings reflect different perspectives among the stakeholders regarding the definition of AI literacy and their priorities regarding the competencies involved when training for AI literacy [2]. In other words, the plurality of perspectives is depicted in the teams' outcomes. This may indicate that hackathons allow expressions of diversity while at the same time foster common understanding.

Regarding the participants' satisfaction with the hackathon and their continuation intentions (RQ 2), participants were overall positive, although they wished for more time so that they could achieve more and have time for in depth discussions. Still, the majority of participants did not intend to continue with their projects due to other responsibilities, lack of time, or difficulties connecting with team members. This indicates the need for stable structures, opportunities, and incentives for continuation. Continuation after the hackathon is an established research direction [11], indicating that continuation after hackathons is low in general even with participants intending to continue [4].

5.1 Theoretical and Practical Implications

We envision that our work has several implications for research and practice regarding the use of hackathons as participatory forums for facilitating critical discussions. We found that the heterogeneity of the teams was beneficial to the teams' brainstorming process and team members benefited from the diverse input of their teammates. For future events, this could be replicated through guided group formation or event participant selection [4].

The hackathon was overall well received, based on participation numbers, retention, satisfaction, and feedback. However, participants expressed that they would have liked to have more time for *discussion* and to *deliver* finished products. To address this, we suggest that hackathons of this format can be used iteratively to define, communicate and align goals while preparing for prototyping. This setup could for example be utilized as a kick-start brainstorming event for a collaboration, either internally or including external participants [6]. This would remove the pressure of completing a working artifact within the time constraints as well as the barrier of availability after the event for continuation. To achieve more within the event, the hackathon could be extended to a two-day (or longer) event. This would allow for a full day of ideation and additional days for implementation of the project. With an extended event however come additional considerations about duration and what weekdays to pick as well as availability of prospective participants [1].

6 Conclusion

Our findings show that we can utilize hackathons to bring together participants with varying prior knowledge or expertise of the hackathon theme and have a productive discussion. However, we found that with the event's limited time frame came certain shortcomings, concretely reduced time for discussions in favor of producing an artifact and not enough time to produce complete products.

We acknowledge that this research is subject to limitations. Our focus on launching a prototype of the proposed event resulted in a small hackathon and thus a small data sample that was tightened by the few responses to post-questionnaire and interviews. This limits the generalizability of our findings and could have contributed to an interpretation bias. For further investigation, the organization of additional and potentially larger events is needed.

Based on our findings we suggest refinement of the format for future iterations. Concretely, prioritization of ideation and discussion, and an extension of the time-frame to provide room for creation of products. As such, the goals of the event need to direct the setting and should be communicated clearly.

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